Datasets: 6 tables, 3 source systems, no declared keys

erp.PROJECT ERP, 5,000 rows

ID	PROJECT_CODE	NAME	STATUS	DATA
1	1042	Atlas Migration	2	N/A
2	1043	Beacon Audit	5	TBD
3	2001	Cedar Rollout	1	1
…	…	…	…	…

erp.PROJECT_STG ERP, 812 rows

PROJECT_CODE	NAME
2001	Cedar Rollout
2002	Delta Review
3107	Echo Upgrade
…	…

erp.LEDGER ERP, 120,000 rows

ID	PRJ_ID	LINE_NO	AMOUNT	STATUS
1	1042	1	1250.00	2
2	2001	2	310.40	2
3	3107	1	98.00	4
…	…	…	…	…

ts.TIMESHEET Timesheet, 80,000 rows

EMP_ID	WEEK	ENGAGEMENT	HOURS	DATA
E17	2026-W14	ENG-1042	7.5	ok
E17	2026-W15	ENG-2002	8.0	N/A
E23	2026-W14	ENG-1042	6.0	1
…	…	…	…	…

ts.TS_APPROVAL Timesheet, 79,600 rows

EMP_ID	WEEK	APPROVER
E17	2026-W14	M04
E17	2026-W15	M04
E23	2026-W14	M11
…	…	…

bill.INVOICE_EXP Billing export, 9,300 rows

INV_NO	PROJ	TOTAL
INV-77	0001042	4800.00
INV-78	0002001	1120.50
INV-79	0001042	260.00
…	…	…

① Column Profiler: one pass, no LLM call

	Statistics			Pattern + Normalization		Value Sketches		LLM Annotation + Embedding	
Column	type	$u(c)$	null	pattern	normalized	HLL ndv	MinHash $h_1..h_4$ (of 128)	description, tag	embedding (of 384)
PROJECT.ID	INT	1.00	0%	9{1,4}	=	5,000	[3, 17, 25, 40]	surrogate row id, –	[0.12,-0.40, 0.05,..]
PROJECT.PROJECT_CODE	INT	1.00	0%	9999	=	5,000	[11, 46, 93, 120]	project identifier, **Project**	[0.71, 0.08,-0.22,..]
PROJECT.STATUS	INT	0.001	0%	9	=	5	[3, 17, 81, 95]	project status code, –	[0.02, 0.55, 0.31,..]
PROJECT.DATA	VARCHAR	0.02	31%	mixed	=	96	[3, 11, 17, 46]	free-form note, –	[-0.10, 0.02, 0.04,..]
PROJECT_STG.PROJECT_CODE	INT	1.00	0%	9999	=	812	[11, 46, 93, 131]	project identifier, **Project**	[0.69, 0.10,-0.20,..]
LEDGER.ID	INT	1.00	0%	9{1,6}	=	120,000	[0, 3, 11, 17]	surrogate row id, –	[0.13,-0.38, 0.06,..]
LEDGER.PRJ_ID	INT	0.04	0.2%	9999	=	4,702	[11, 46, 93, 120]	project of the ledger line, **Project**	[0.66, 0.12,-0.18,..]
LEDGER.LINE_NO	INT	0.0003	0%	9{1,2}	=	40	[3, 17, 25, 52]	line number in a posting, –	[0.05, 0.21, 0.40,..]
TIMESHEET.ENGAGEMENT	VARCHAR	0.05	1%	AAA-9999	ENG-1042⟶1042	3,911	[11, 46, 93, 140]	client engagement worked on, **Project**	[0.52, 0.30,-0.11,..]
TIMESHEET.HOURS	DECIMAL	0.001	0%	9.9	=	33	[5, 17, 70, 88]	hours worked, –	[0.01, 0.62, 0.17,..]
TIMESHEET.(EMP_ID,WEEK)	composite	1.00	0%	A99,9999-A99	=	80,000	[2, 9, 30, 61]	employee week, Timesheet	[0.33,-0.07, 0.48,..]
TS_APPROVAL.(EMP_ID,WEEK)	composite	1.00	0%	A99,9999-A99	=	79,600	[2, 9, 30, 61]	approved employee week, Timesheet	[0.31,-0.05, 0.47,..]
INVOICE_EXP.PROJ	VARCHAR	0.44	0%	0009999	0001042⟶1042	4,118	[11, 46, 93, 120]	project billed, **Project**	[0.58, 0.04,-0.25,..]

(1a) Statistics from one warehouse scan per batch of columns; $u(c)$ = distinct / rows.

(1b) Normalization strips the prefix ENG- and the leading zeros of 0001042, so both columns are sketched again on normalized values.

(1c) Sketches: HLL for ndv; MinHash for containment. Equal minima [11, 46, 93, ..] already show that the project-code columns overlap.

(1d) Annotation is reused from the pipeline; the embedding encodes the column name and description, then the table name and description.

② Ring Builder: 3 candidate rings

(2a) Key Candidates: $u \geq 0.95$, nulls $\leq 5\%$

PROJECT.ID	5,000
PROJECT.PROJECT_CODE	5,000
PROJECT_STG.PROJECT_CODE	812
LEDGER.ID	120,000
TIMESHEET.(EMP_ID,WEEK)	80,000 composite
INVOICE_EXP.INV_NO	9,300

(2b) Candidate Index: LSH Ensemble on normalized MinHash ($\hat{c}$) and ANN on embeddings (cos)

referencing column	⟶ candidate owner	$\hat{c}$	cos	
LEDGER.PRJ_ID	⟶ PROJECT.PROJECT_CODE	0.99	0.88	
LEDGER.PRJ_ID	⟶ PROJECT.ID	1.00	0.21	accidental: 1042 ∈ 1..5000
LEDGER.PRJ_ID	⟶ LEDGER.ID	1.00	0.18	accidental
PROJECT_STG.PROJECT_CODE	⟶ PROJECT.PROJECT_CODE	1.00	0.97	staging copy
INVOICE_EXP.PROJ	⟶ PROJECT.PROJECT_CODE	0.97	0.74	raw: $\hat{c} = 0.00$
TIMESHEET.ENGAGEMENT	⟶ PROJECT.PROJECT_CODE	0.95	0.69	raw: $\hat{c} = 0.00$
LEDGER.LINE_NO	⟶ PROJECT.ID	1.00	0.09	accidental: 1..40
PROJECT.STATUS	⟶ PROJECT.ID	1.00	0.05	accidental: 1..5
TIMESHEET.HOURS	⟶ PROJECT.ID	0.40	0.02	
PROJECT.DATA	⟶ 38% of all key candidates			
TS_APPROVAL.(EMP_ID,WEEK)	⟶ TIMESHEET.(EMP_ID,WEEK)	0.99	0.92	composite

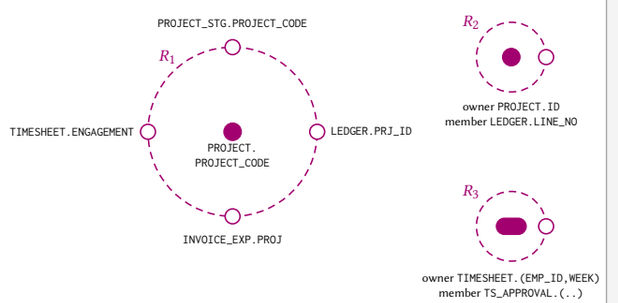
(2c) Scope + Prune

Scope: the concept tag **Project** spans ERP, Timesheet, and Billing, so candidates are kept *across* source systems.

Prune:
× PROJECT.DATA: generic, matches 38% of key candidates
× PROJECT.STATUS: code with 5 values
× TIMESHEET.HOURS: DECIMAL, type-incompatible with INT owners

Owner election: LEDGER.PRJ_ID joins the ring of its best-scoring owner, PROJECT_CODE ($\hat{c}$ 0.99, cos 0.88), not PROJECT.ID or LEDGER.ID. The ring of the staging copy merges into it; the owner is the larger domain (5,000 > 812).

(2d) Candidate Rings



③ Budgeted Verifier: 1 LLM call for 6 candidates

(3a) Calibration + Optimizer

Calibration: all-pairs LLM labels on a random sample of 60 candidates of the whole database: 14 joins.

Optimizer: recall target $\gamma = 0.95$, confidence $1 - \delta = 0.95$ ⇒ reject below $\tau_{lo} = 0.20$, accept above $\tau_{hi} = 0.90$ ⇒ at most 8 candidates per prompt.

(3b) Distilled Model: score from profile features

member	⟶ owner	score	decision
PROJECT_STG.PROJECT_CODE	⟶ PROJECT_CODE	0.97	accept
LEDGER.PRJ_ID	⟶ PROJECT_CODE	0.93	accept
INVOICE_EXP.PROJ	⟶ PROJECT_CODE	0.71	uncertain
TIMESHEET.ENGAGEMENT	⟶ PROJECT_CODE	0.46	uncertain
LEDGER.LINE_NO	⟶ PROJECT.ID	0.06	reject
TS_APPROVAL.(EMP_ID,WEEK)	⟶ TIMESHEET.(..)	0.94	accept

Verdict cache: R_1 has uncertain members and no cached verdict ⇒ 1 prompt. R_2 loses its only member ⇒ dropped. R_3 has none uncertain ⇒ no prompt.

(3c) Ring Prompt for R_1: one LLM call

```
Owner: erp.PROJECT.PROJECT_CODE (INT, unique, 5,000 values)
  "project identifier"; e.g. 1042, 1043, 2001
Accepted members: PROJECT_STG.PROJECT_CODE, LEDGER.PRJ_ID
Uncertain candidates:
  1. bill.INVOICE_EXP.PROJ "project billed"; e.g. 0001042, 0002001
     normalized 1042, 2001; containment 0.97
  2. ts.TIMESHEET.ENGAGEMENT "client engagement worked on";
     e.g. ENG-1042, ENG-2002; normalized 1042, 2002; containment 0.95
Is the owner a business key? Which candidates refer to it?
```

LLM answer: {owner: confirmed, members: [1, 2]}

Both answers become labels for the distilled model (cold start).

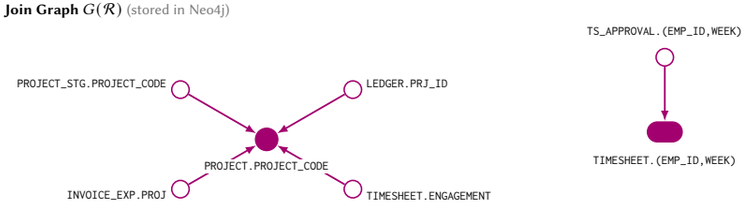
(3d) SQL Join Check: exact containment of members accepted by the distilled model

PROJECT_STG.PROJECT_CODE	1.000 ✓
LEDGER.PRJ_ID	0.998 ✓
TS_APPROVAL.(EMP_ID,WEEK)	1.000 ✓

(3e) Transitive Propagation: the 4 confirmed members of R_1 give 4 member⟶owner joins, and $\binom{4}{2} = 6$ member↔member joins follow without an LLM call, e.g., LEDGER.PRJ_ID = norm(INVOICE_EXP.PROJ).

Output

Join Graph $G(\mathcal{R})$ (stored in Neo4j)



5 edges, a star per key domain; no edge to PROJECT.ID, LEDGER.ID, STATUS, or DATA.

Cost

All-pairs prompting: $\binom{6}{2} = 15$ prompts for these 6 tables, each with two schemas and sample rows.

KeyRing: 1 ring prompt, plus the calibration sample, which is shared by the whole database.

Next day: incremental update

A new table fin.BUDGET arrives with PROJECT_NO (1042, 3107, ..). Its profile is sketched, the index finds owner PROJECT_CODE ($\hat{c}$ 0.98, cos 0.86), and it joins R_1 without a rebuild.